

## LAB EXPERIMENT RECORD

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| <b>Department</b>                | AIDS                                      |
| <b>Course Code / Course Name</b> | DSA0601 / Data Handling and Visualization |
| <b>Experiment</b>                | Data Visualization using R                |
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| <b>Register Number</b>           | 192424055                                 |
| <b>Date</b>                      | 03 October 2026                           |
| <b>Day</b>                       | Saturday                                  |

### SET 7 — Customer Demographics Analysis

#### Aim

To analyze customer demographic data using **bar charts, pie charts, tables, and a Tableau dashboard.**

#### Dataset

| <b>Customer ID</b> | <b>Age</b> | <b>Gender</b> | <b>Income (\$)</b> |
|--------------------|------------|---------------|--------------------|
| 1                  | 28         | Female        | 50,000             |
| 2                  | 35         | Male          | 60,000             |
| 3                  | 42         | Female        | 75,000             |

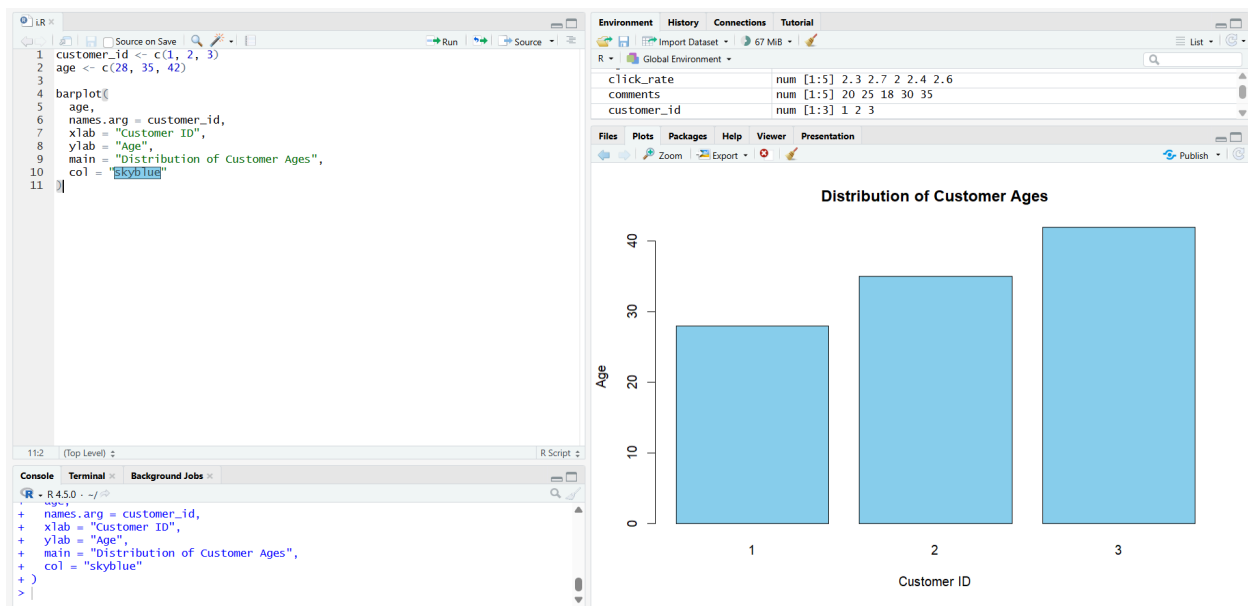
## Q1. Bar Chart — Distribution of Customer Ages

### R Program

```
customer_id <- c(1, 2, 3)
age <- c(28, 35, 42)

barplot(
  age,
  names.arg = customer_id,
  xlab = "Customer ID",
  ylab = "Age",
  main = "Distribution of Customer Ages",
  col = "skyblue"
)
```

### Output



### Observation

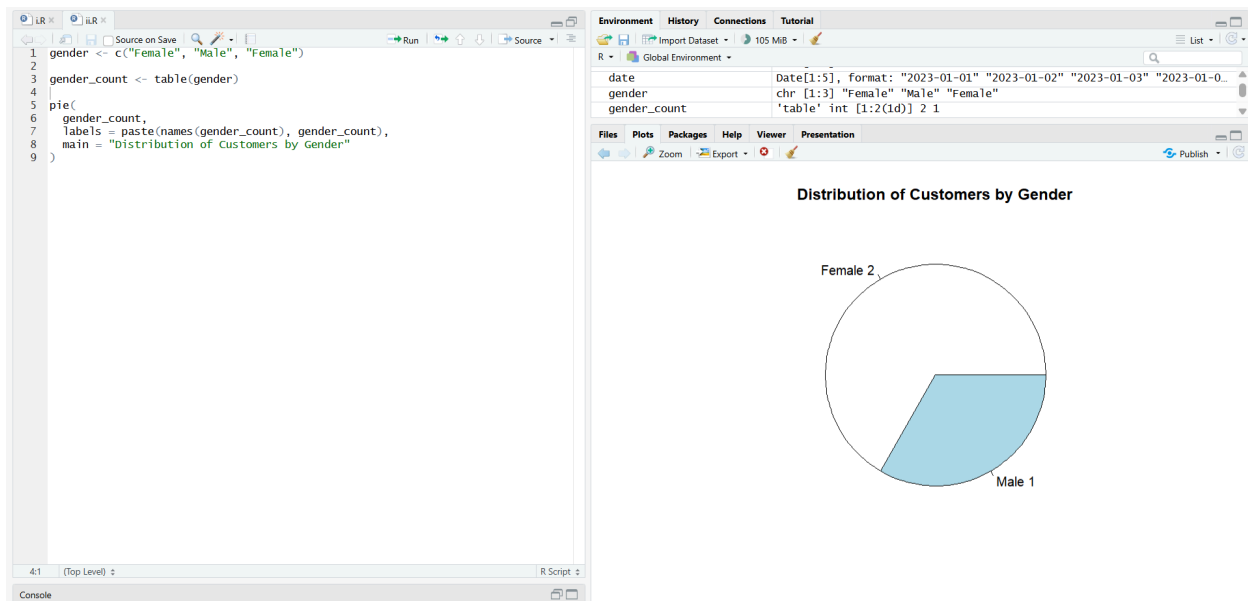
Female customers represent **2 out of 3 customers**, while male customers represent **1 out of 3**.

## Q2. Pie Chart — Customers by Gender

### R Program

```
gender <- c("Female", "Male", "Female")
gender_count <- table(gender)
pie(
  gender_count,
  labels = paste(names(gender_count), gender_count),
  main = "Distribution of Customers by Gender"
)
```

### Output



### Observation

Female customers represent **2 out of 3 customers**, while male customers represent **1 out of 3**.

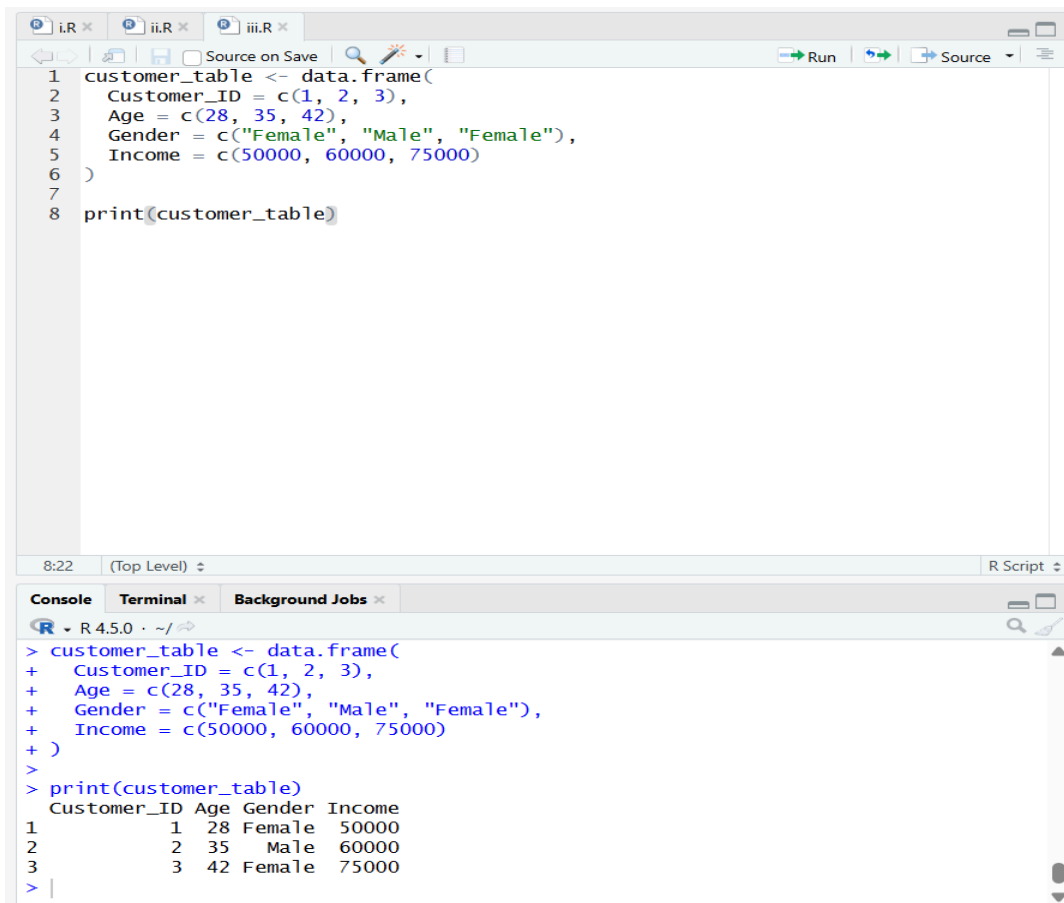
### Q3. Demographic Information Table

#### R Program

```
customer_table <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)
```

```
print(customer_table)
```

#### Output



The screenshot shows an R Studio interface with three tabs at the top: 'i.R x', 'ii.R x', and 'iii.R x'. The 'iii.R x' tab is active, displaying the following R code:

```
1 customer_table <- data.frame(  
2   Customer_ID = c(1, 2, 3),  
3   Age = c(28, 35, 42),  
4   Gender = c("Female", "Male", "Female"),  
5   Income = c(50000, 60000, 75000)  
6 )  
7  
8 print(customer_table)
```

Below the code editor, the 'Console' tab is active, showing the execution of the code and the resulting output:

```
> customer_table <- data.frame(  
+   Customer_ID = c(1, 2, 3),  
+   Age = c(28, 35, 42),  
+   Gender = c("Female", "Male", "Female"),  
+   Income = c(50000, 60000, 75000)  
+ )  
>  
> print(customer_table)  
  Customer_ID Age Gender Income  
1           1  28 Female 50000  
2           2  35  Male 60000  
3           3  42 Female 75000  
>
```

#### Observation

The table provides the complete demographic information for each customer.

## Q4. Tableau Dashboard

### Objective

Create an interactive dashboard combining:

1. **Bar chart** — Customer Age
2. **Pie chart** — Gender Distribution
3. **Table** — Customer Demographic Information

#### Step 1 — Prepare Excel Data

Create an Excel file with:

| Customer ID | Age | Gender | Income |
|-------------|-----|--------|--------|
| 1           | 28  | Female | 50000  |
| 2           | 35  | Male   | 60000  |
| 3           | 42  | Female | 75000  |

Save it as:

*customer\_demographics.xlsx*

#### Step 2 — Import into Tableau

1. Open **Tableau**.
2. Select **Microsoft Excel**.
3. Select *customer\_demographics.xlsx*.
4. Open the worksheet.

#### Step 3 — Create Bar Chart

- Drag **Customer ID** → Columns.
- Drag **Age** → Rows.
- Select **Bar** from the Marks card.
- Add appropriate axis title.

#### Step 4 — Create Pie Chart

- Create a new worksheet.

- Drag **Gender** → Color.
- Drag **Customer ID** → Angle.
- Change Marks type to **Pie**.
- Add Customer ID to **Label** if required.

### Step 5 — Create Demographic Table

- Create another worksheet.
- Drag **Customer ID**, **Age**, **Gender**, and **Income** into the table.
- Format the headings clearly.

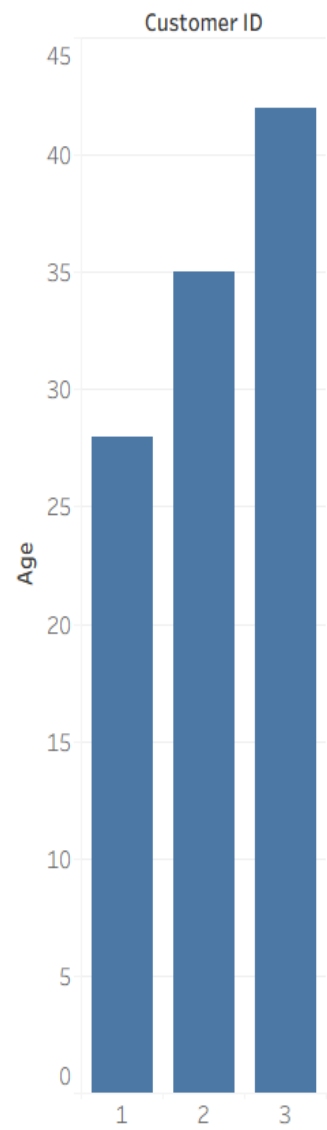
### Step 6 — Create Dashboard

1. Select **New Dashboard**.
2. Drag the **Age Bar Chart** onto the dashboard.
3. Drag the **Gender Pie Chart** onto the dashboard.
4. Drag the **Demographic Table** onto the dashboard.
5. Add filters such as:
  - Gender
  - Age
  - Income
6. Arrange and format the dashboard.

Output

|         |             |
|---------|-------------|
| Columns | Customer ID |
| Rows    | SUM(Age)    |

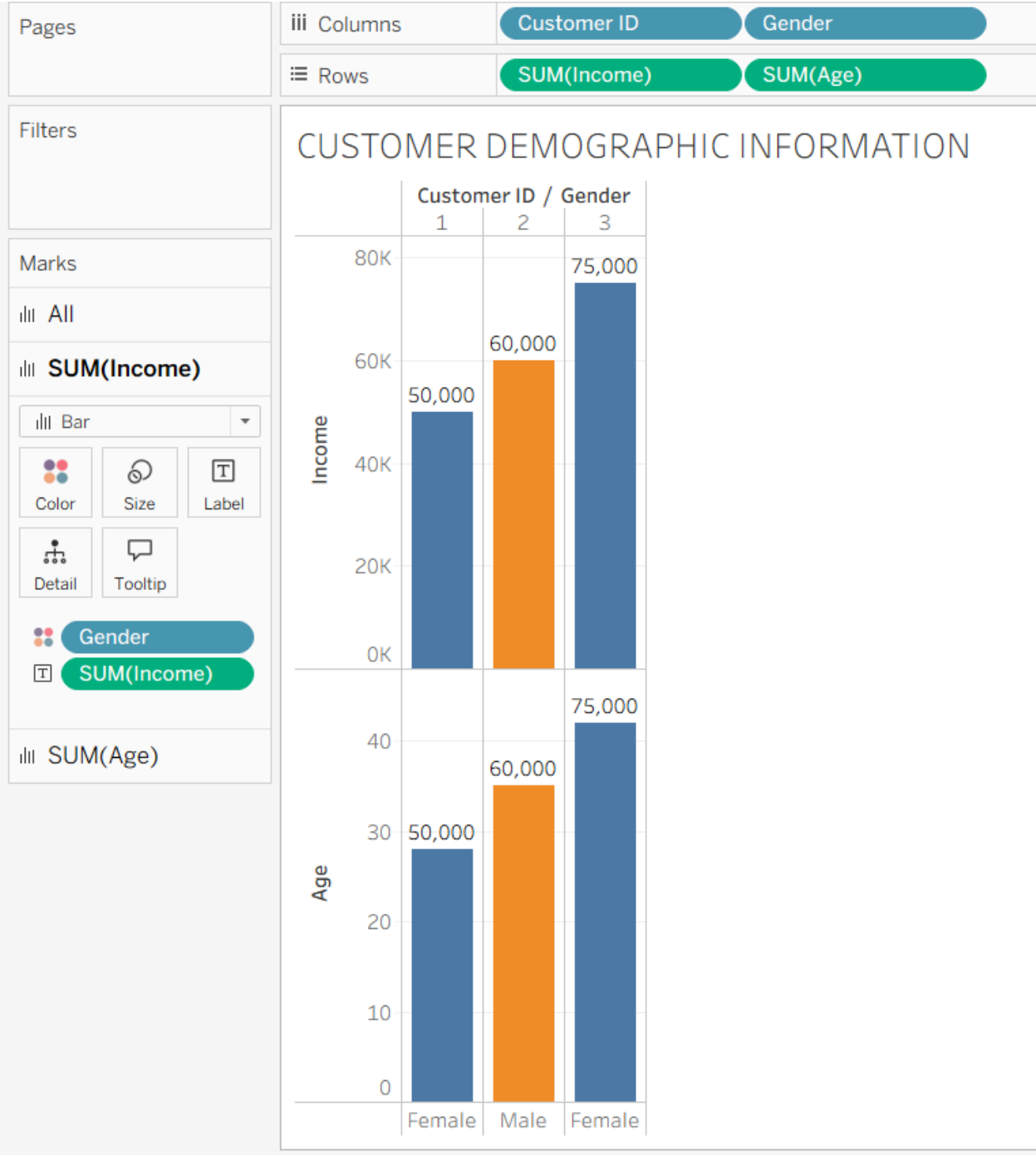
AGE BAR CHART

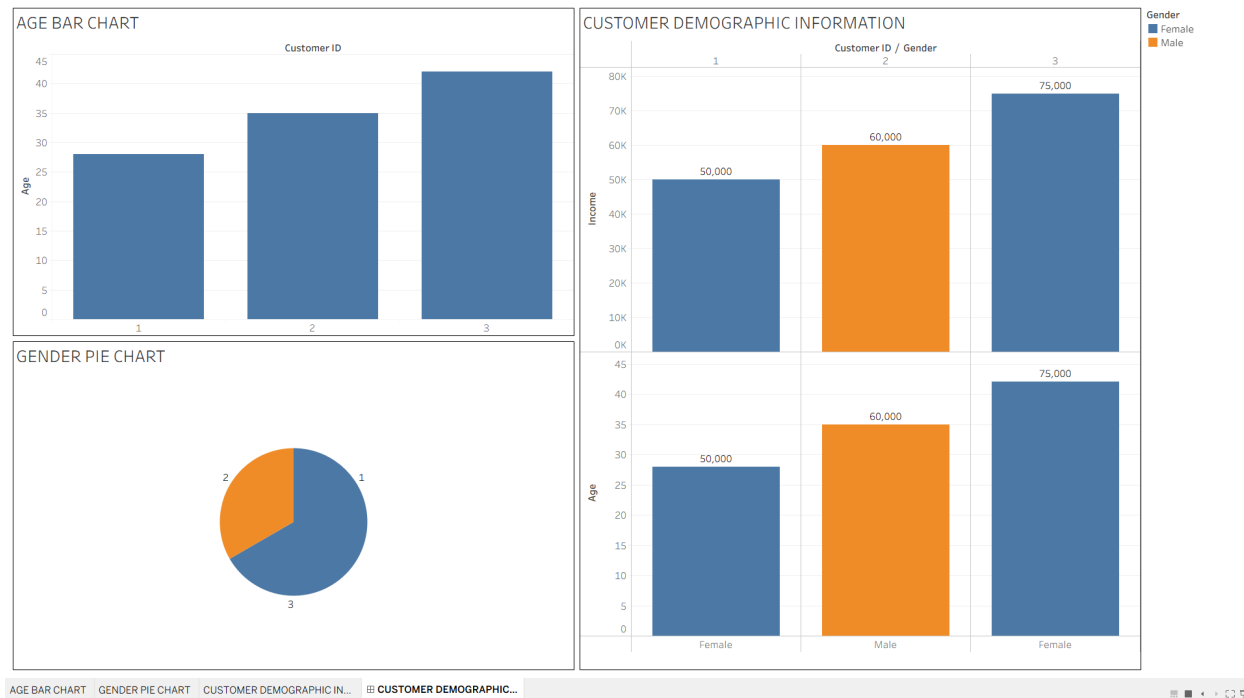


|         |  |
|---------|--|
| Columns |  |
| Rows    |  |

GENDER PIE CHART







## Procedure

1. Open **RStudio** and create a new **R Script**.
2. Enter the given customer demographic data into R.
3. Use the `barplot()` function to create a **bar chart showing the distribution of customer ages** and label the axes and title.
4. Use the `table()` and `pie()` functions to calculate and display the **distribution of customers by gender**.
5. Use the `data.frame()` function to create a **table containing Customer ID, Age, Gender, and Income**.
6. Open **Tableau** and import the customer demographic dataset.
7. Create a **bar chart for customer ages**, a **pie chart for gender distribution**, and a **demographic information table**.
8. Combine the three visualizations into a single **interactive Tableau dashboard**.
9. Add suitable labels, titles, and filters for easier exploration of the demographic data.

## Result

The customer demographic dataset was successfully analyzed using **R and Tableau**. A **bar chart** was created to visualize customer ages, a **pie chart** was created to show gender distribution, and a **demographic table** was generated. These visualizations were successfully combined into an **interactive Tableau dashboard** for exploring customer demographics.